11D — Deep-Dive Feature Audit: Human Foundation, Psychology, Social Graph & Karma (Part 4 of 5)

Social Graph (`rel`), Unused Attachment Styles, Missing Kinship/Celibacy Guards in Romance, Split-Brain `spouse` vs `companionId`, Dormant Multi-Gen Offspring & Three-Way Mentorship Desync

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Immutable Rule**: **NEVER modify `js/human_core.js`** (`155:2f9c9dd7`). All fixes operate in `js/ai.js`, `js/sect.js`, `js/cognition.js`, `js/combat.js`, and `js/sim.js`.

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1. `[HUM-SOC-01]` Critical Missing Kinship, Generational & Celibacy-Precept Guards in Romance & Marriage (`js/ai.js:684, 748–757, 786`)!

1.1 Verbatim Code Proof (`js/ai.js:684` & `786`)

JavaScript

```
// AI.socialStart (line 786):
if (n.companionId === o.id || (r.l || 0) > 25 && E(n, 'interests.romance') > 45 ||
((r.a || 0) > 45 && E(n, 'interests.romance') > 60 && E(o, 'interests.romance') >
45 && Math.abs(n.h.birthDay - o.h.birthDay) < 360 * 14))
  opts.push({ t: 'date', ... });

// AI.updateBond (lines 684, 697):
if (r.l > 62 && r.a > 40) nb = r.l > 85 && r.a > 70 ? 'คู่ครอง' : 'คนรัก';
...
if (nb === 'คู่ครอง' && !x.h.life.family.spouse && !other.h.life.family.spouse) {
x.h.life.family.spouse = other.id; ... }
```

1.2 Four Severe Contextual & Ethical Bugs:

1. **Zero Consanguinity / Kinship Filter (`Incest Hazard Once Multi-Gen Offspring Are Enabled!`)**:
   - Notice `(r.l || 0) > 25` bypasses the `14`-year age check, AND **brother-sister siblings (`sharing parentIds / family.parents`!) ARE BORN WITHIN 14 YEARS OF EACH OTHER WITH HIGH AFFECTION `r.a > 45`!**
   - Without an explicit `SECT.areBloodRelatives(a, b)` check (`checking parents, grandparents, children, and shared parentIds up to 3 degrees`), siblings—or a widowed parent and adult child—can roll `'date'`, build `r.l > 85`, and **marry their own blood sibling or parent**!
2. **Zero Celibate / Ascetic Vow Guard (`VOW_ASCETIC_PRECEPT` & Monastic `BUDDHA_DHARMA`)**:
   - Sworn celibate monks (`voluntaryVows.VOW_ASCETIC_PRECEPT.active` or monastic Buddhist Arhats) freely go on romantic dates and marry without hesitation or vow violation!
3. **Polygamous Lover Drift & Dead Spouse Pointer Lock**:
   - Notice `AI.socialStart` (`line 786`) does **NOT** check `!n.h.life.family.spouse && !o.h.life.family.spouse` before pushing `{ t: 'date' }`! A married Elder (`with a living spouse!`) still invites random 20-year-old disciples on romantic `'date'` sessions at `spirit_pond` whenever `r.a > 45 && interests.romance > 60`, turning half the sect into extra-marital `'คนรัก'`!
   - Conversely, in `AI.kill` (`ai.js:848`), when `spouse` dies, `o.h.life.family.spouse` still points to the dead ID forever (`without distinguishing devoted eternal widowhood vs eventual remarriage`).
4. **Generated `h.attachment` (`มั่นคง / วิตกกังวล / หลีกเลี่ยง / สับสน`) Is `100%` Unused (`0` Reads!)**:
   - `human_core.js:293` computes attachment style from `trustfulness + empathy + composure`, yet `ai.js` never reads `n.h.attachment` in `acceptProb`, `date`, jealousy, or heartbreak!

- **Exact Remediation Blueprint**:
  - Add **`SECT.canRomance(W, a, b)`** checking:
    1) `!SECT.areBloodRelatives(W, a, b)` (`parents, children, siblings, ancestor/descendant up to 3 generations` — **HARD BLOCK**),
    2) `HC.ageYears(a) >= 17 && HC.ageYears(b) >= 17`,
    3) Neither has a **living** `spouse` or `companionId` (`unless partner === other`),

4) Neither holds an active celibacy vow (`VOW_ASCETIC_PRECEPT` / `daoOath === 'CELIBATE_MONK'`),
5) Weight acceptance and jealousy by `h.attachment` (`'หลีกเลี่ยง'` resists rapid commitment; `'วิตกกังวล'` suffers stronger heartbreak/jealousy).

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2. `[HUM-SOC-02]` Split-Brain Marriage (`family.spouse`) vs Taoist Companion (`companionId`) & `0` Natural Births Across Multi-Century Dynasties (`js/ai.js:697, 738` vs `js/sect.js`)!

1. **Organic Marriages Receive `0%` of Taoist-Companion Gameplay Mechanics**:
   - When two cultivators fall in love and marry organically in `AI.updateBond` (`ai.js:697`: `x.h.life.family.spouse = other.id`), **`x.companionId` is NEVER set**!
   - Yet **every gameplay synergy in the engine checks `companionId`, NOT `family.spouse`**:
     - Dual-cultivation `+1.2 Qi` & `-0.5 heartDemon` in `AI.pairTick` (`ai.js:738`)
     - Twin-Lotus Duet Skill `DUET_TWIN_LOTUS_SWORD` (`combat.js` / `v340:89`)
     - Bodyguard interception priority (`combat.js:409`)
     - Bereavement `+15 heartDemon` & grief (`ai.js:850`)
   - **Fix**: Whenever two cultivators marry (`nb === 'คู่ครอง'`), automatically call `SECT.bindTaoistCcompanions(W, x, b)` (`setting companionId, r.bond = 'TAOIST_COMPANION', and RED_THREAD`)!
2. **Zero Autonomous Multi-Generational Births (`SECT.spawnCompanionOffspring` Is 100% Player-Button Locked!)**:
   - Although `v3.6.0` (`PROP-34`) built 3-generation clan lineages (`parentIds`, `+30%` ancestral technique study speed, atavistic `TR_BL_*` awakening), **no married couple ever has a child autonomously**!
   - **Emergent Lifecycle Hook**: In `SIM.monthly`, when a married heterosexual couple (`both alive, in-sect, !away, !detained, bioAge 18..48, housingCapacity has room, cooldown >= 5 years, < 2–3 children based on values.familyClan`) shares high affection (`r.a >= 75`), allow a natural `4%–8%/month` chance to welcome a child (`inheriting blended latent genome` lat, `family surname` clan, `heritable` CONST_/*TR_BL*_`traits, and setting`parents: [father.id, mother.id]` cleanly)!

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3. [HUM-SOC-03] Three-Way Mentorship Schema Split & Missing `killer` Parameter in `AI.kill` (`js/ai.js:833–865`, `js/sim.js:244`, `js/sect.js`)!

1. **Three Unsynchronized Mentorship Fields**:
   - Founding (`sim.js:244`), Tournament (`combat.js:843`), and Personal Disciple (`sect.js`) write `d.mentor = m.id` and `m.h.life.family.disciples`, **while `AI.kill` (`ai.js:835`), `AI.pairTick` (`line 790`), and `DUET_MASTER_DISCIPLE_INHERITANCE` check `m.apprenticeIds` and `d.mentorId`**!
   - **Fix**: Route every mentorship assignment through a single **`SECT.MENTORSHIP.formBond(W, masterId, discipleId)`** helper that writes `d.mentor = d.mentorId = d.h.life.family.master = masterId` and pushes `discipleId` into both `m.apprenticeIds` and `m.h.life.family.disciples`!
2. **`AI.kill(W, n, why)` Drops the `killer` Entity — Breaking Blood-Vengeance Chains (`ai.js:833` vs `combat.js:687`)!**:
   - `COMBAT.slay` calls `AI.kill(W, d, 'ถูก ' + C.unitName(a) + ' สังหาร')` **without passing `{ killer: a, killerSectId: a.sectId }`**!
   - Consequently, when a rival sect champion or traitor slays an NPC's **Master, Taoist Companion, Parent, or Child**, `AI.kill` inflicts `grief`—**but NEVER records the killer/rival-sect in `o.h.mind.grudges`, NEVER binds `BLACK_THREAD` against a living killer, and NEVER awards `TR_AC_MASTER_VENGEANCE` / `TR_AC_HEART_DEMON_LOVE`**! Pass `opts = { killer, killerSectId }` into `AI.kill`!

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4. [HUM-SOC-04] Unbounded `O(N²)` Dead-ID Accumulation in `h.mind.relations` Across Multi-Century Runs

- Over 200+ years (`hundreds of buried/retired/exiled NPCs`), every living NPC's `h.mind.relations` retains entries for long-dead strangers (`r.n <= 2, no bond, not family/mentor`), bloating `SIM.serialize(W)`.
- **Safe Pruning Rule (`SIM.monthly`)**: Prune `relations[deadId]` **only** when `deadId` is in `W.graveyard` / `W.gone`, has **no** `bond`, is **not** in `family` (`parents/spouse/children/master/disciples`), and `Math.abs(r.a) < 35 && (r.v || 0) < 35`.